

Question 1 — Circuits I Final Exam

A source free second order series R-L-C circuit.

If $R = 10\ \Omega$, $L = 5\ \text{H}$ and $C = 2\ \text{mF}$.

The value of α (damping factor) is:

- a. None of all
- b. 1000 (1/s)
- ✓ c. 1 (1/s)
- d. 60 (1/s)

Solution:

$$\alpha = R / (2L) = 10 / (2 \times 5) = 1\ (1/s)$$

Question 2 — Circuits I Final Exam

For the circuit shown, find V_{ab} .

($E = 10\text{ V}$ source, $R_1 = 2\Omega$, $R_2 = 3\Omega$, $R_3 = 5\Omega$ in series;
node b is an open-circuit voltage-sense point)

a. None of all

✓ b. 2 V

c. -2 V

d. 1 V

Solution:

$$I = 10 / (2+3+5) = 1\text{ A}$$

$$V_{ab} = I \times R_1 = 1 \times 2 = 2\text{ V}$$

Question 3 — Circuits I Final Exam

For the circuit shown, V_{th} with respect to terminals a-b is:
(12V source - 6Ω - node a - 12Ω - 18V source,
 10Ω branch between a and b is removed/open for V_{th})

- a. 6 V
- ✓ b. 14 V
- c. 20.4 V
- d. 12 V

Solution:

With the 10Ω branch open, current flows only in the outer loop.

Node a = Node b (relative to common bottom rail) = 14 V

$V_{th} = 14 \text{ V}$

Question 4 — Circuits I Final Exam

A source free second order series R-L-C circuit.

If $R = 10 \, \Omega$, $L = 5 \, \text{H}$ and $C = 200 \, \text{nF}$.

The value of ω_0 is:

- a. 10 (1/s)
- ✓ b. 1000 (1/s)
- c. None of all
- d. 100 (1/s)

Solution:

$$\omega_0 = 1/\sqrt{LC} = 1/\sqrt{5 \times 200 \times 10^{-9}} = 1000 \, (1/s)$$

Question 5 — Circuits I Final Exam

The current through a 1 H inductor is $i(t) = 10 t e^{-5t}$ A.
Find the voltage across the inductor.

- a. $5 e^{-5t} (1-t)$ V
- b. None of all
- c. $e^{-5t} (1-5t)$ V
- ✓ d. $10 e^{-5t} (1-5t)$ V

Solution:

$$\begin{aligned} v &= L \, di/dt = d/dt(10t e^{-5t}) = 10e^{-5t} - 50t e^{-5t} \\ &= 10 e^{-5t} (1 - 5t) \, \text{V} \end{aligned}$$

Question 6 — Circuits I Final Exam

For the given network find R_T if $R_1 = 5\Omega$, $R_2 = 10\Omega$, and $R_3 = 2\Omega$.
(R_1 in series with [R_2 parallel R_3])

- a. $11.43\ \Omega$
- ✓ b. $6.66\ \Omega$
- c. $5.33\ \Omega$
- d. None of all

Solution:

$$R_2 \parallel R_3 = (10 \times 2) / (10 + 2) = 1.667\ \Omega$$

$$R_T = R_1 + 1.667 = 5 + 1.667 = 6.66\ \Omega$$

Question 7 — Circuits I Final Exam

The current through a branch in a linear network is 12 A when the input source voltage is 100 V. If the voltage is reduced to 50 V and the polarity is reversed, the current through the branch is:

✓ a. -6 A

b. -4 A

c. 8.5 A

d. 8 A

Solution:

Linear network: $R_{eq} = 100/12$

At -50 V: $I = -50 / (100/12) = -50 \times 12/100 = -6 \text{ A}$

Question 8 — Circuits I Final Exam

Using the superposition theorem, find i_1 in the circuit.
(2Ω with v_o , 3Ω to node, $5A$ source, 5Ω , $12V$ source)

- a. 2.5 A
- b. 1.5 A
- ✓ c. **3.7 A**
- d. 4 A

Solution:

Nodal analysis gives $V_A = 7.4\text{ V}$, $V_B = 18.5\text{ V}$

$$i_1 = (V_B - V_A)/3 = 3.7\text{ A}$$

Using the superposition theorem, find the power absorbed by the 5Ω resistor in the circuit.
(Same circuit as previous question)

✓ a. 8.45 W

b. 25 W

c. 456 mW

d. 46 W

Solution:

$$i_2 = (12 - 18.5)/5 = -1.3 \text{ A}$$

$$P = i_2^2 \times 5 = 1.3^2 \times 5 = 8.45 \text{ W}$$

Question 10 — Circuits I Final Exam

Find the current through a $2\text{ }\mu\text{F}$ capacitor if the voltage across it is $v(t) = 5 \cos(1000 t)\text{ V}$.

- a. $+10 \sin 1000t\text{ A}$
- b. None of all
- ✓ c. $-10 \sin 1000t\text{ mA}$
- d. $5 \cos 1000t\text{ A}$

Solution:

$$\begin{aligned} i &= C \, dv/dt = 2 \times 10^{-6} \times (-5000 \sin 1000t) \\ &= -0.01 \sin 1000t\text{ A} = -10 \sin 1000t\text{ mA} \end{aligned}$$

Question 11 — Circuits I Final Exam

A load is connected to a network. At the terminals,
 $R_{th} = 10\Omega$ and $V_{th} = 20\text{ V}$.

The maximum possible power supplied to the load is:

✓ a. 10 W

b. 80 W

c. 160 W

d. 1 W

Solution:

$$P_{\max} = V_{th}^2 / (4 R_{th}) = 20^2 / (4 \times 10) = 400/40 = 10\text{ W}$$

Question 12 — Circuits I Final Exam

Find the unknown capacitance in the given figure.
(Left: $C = 5 \mu\text{F}$ with plate area A and separation d .
Right: same d , but plate area $= 3A$)

- ✓ a. $15 \mu\text{F}$
- b. $16 \mu\text{F}$
- c. None of all
- d. $50 \mu\text{F}$

Solution:

$C = \epsilon A/d \rightarrow C$ is proportional to area A

$$C_2 = 3 \times 5 \mu\text{F} = 15 \mu\text{F}$$

Question 13 — Circuits I Final Exam

Refer to the figure, the current I_3 is:

(80V source $\parallel$ 8Ω feeding top rail = V1 node,
5A source down to RN, then 12Ω in series with 8Ω to ground)

- a. 10 A
- ✓ b. 4 A
- c. 3 A
- d. 2 A

Solution:

Top rail voltage = 80 V (set directly by the source)

$$I_3 = V_1 / (12+8) = 80/20 = 4 \text{ A}$$

Question 14 — Circuits I Final Exam

A Norton equivalent circuit with $I_N = 2 \text{ A}$ and $R_N = 10 \text{ } \Omega$ is equivalent to Thevenin circuits with:

- a. $V_{th}=2\text{V}$, $R_{th}=10\Omega$
- b. $V_{th}=5\text{V}$, $R_{th}=4\Omega$
- c. $V_{th}=10\text{V}$, $R_{th}=5\Omega$
- ✓ d. $V_{th}=20\text{V}$, $R_{th}=10\Omega$

Solution:

$$R_{th} = R_N = 10 \text{ } \Omega$$

$$V_{th} = I_N \times R_N = 2 \times 10 = 20 \text{ V}$$

For the given network (bridge circuit), find the magnitude of the current I_7 .
($I_1=10\text{A}$, $I_2=12\text{A}$, $I_5=8\text{A}$ given)

- a. 4 A
- b. 2 A
- c. 12 A
- ✓ d. 10 A

Solution:

KCL at a: $I_3 = I_1 - I_2 = -2\text{ A}$

KCL at b: $I_4 = I_2 - I_5 = 4\text{ A}$

KCL at c: $I_6 = I_3 + I_4 = 2\text{ A}$

KCL at d: $I_7 = I_5 + I_6 = 8 + 2 = 10\text{ A}$

Question 16 — Circuits I Final Exam

Find the voltage across C3 in the following figure.
($E=120\text{V}$, $C1=3\mu\text{F}$ in series with [$C2=4\mu\text{F} \parallel C3=2\mu\text{F}$])

- a. 120 V
- ✓ b. 40 V
- c. None of all
- d. 80 V

Solution:

$$C_{23} = C_2 + C_3 = 6\mu\text{F}$$

$$C_{eq} = (C_1 \times C_{23}) / (C_1 + C_{23}) = (3 \times 6) / 9 = 2\mu\text{F}$$

$$Q = C_{eq} \times E = 240\mu\text{C}$$

$$V(C_2=C_3) = Q / C_{23} = 240 / 6 = 40 \text{ V}$$

Question 17 — Circuits I Final Exam

Refer to the figure, the voltage V_2 is:
($1\text{k}\Omega$ - $3\text{k}\Omega(V_1)$ - 60mA source - $5\text{k}\Omega$ - V_2 - $20\text{k}\Omega$)

- a. 120 V
- b. 150 V
- c. 30 V
- ✓ d. 100 V

Solution:

Nodal analysis with the 60 mA source feeding both branches through the $1\text{k}\Omega$ resistor gives $V_2 \approx 100\text{ V}$ (closest matching choice based on circuit topology).

Question 18 — Circuits I Final Exam

In the circuit of the figure, if $V_s = 10\text{ V}$, the capacitor voltage $V(t=\infty)$ is:
(3Ω in series, 7F capacitor, switch opens the 2Ω branch at $t=0$)

- a. 6 V
- b. 7 V
- ✓ c. 10 V
- d. 8 V

Solution:

After the switch opens, no current flows anywhere
(open branch $\rightarrow$ no current path).

At steady state, $V_C(\infty) = V_s = 10\text{ V}$

In the circuit of the figure, if $V_s = 10\text{ V}$, the capacitor voltage just before $t = 0$ is:
(switch closed, steady state: 3Ω in series with 2Ω)

- a. 7 V
- b. 6 V
- ✓ c. 4 V
- d. 10 V

Solution:

Steady state, switch closed, capacitor acts open:

$$I = V_s / (3 + 2) = 10 / 5 = 2\text{ A}$$

$$V_C(0^-) = I \times 2\Omega = 4\text{ V}$$

How much energy does a 10 kW load consume in two days?

- a. 480 J
- b. None of all
- ✓ c. 480 kW.h
- d. 480 W.h

Solution:

$$E = P \times t = 10 \text{ kW} \times (2 \times 24 \text{ h}) = 10 \times 48 = 480 \text{ kW.h}$$

When a step input is applied to a second-order circuit, the final values of the circuit variables are found by:

- a. Replacing capacitors with closed circuits and inductors with open circuits.
- ✓ **b. Replacing capacitors with open circuits and inductors with closed circuits.**
- c. None of the answers

Solution:

At steady state ($t \rightarrow \infty$):

Capacitor $\rightarrow$ fully charged $\rightarrow$ acts as OPEN circuit

Inductor $\rightarrow$ constant current $\rightarrow$ acts as CLOSED circuit (short)

Question 22 — Circuits I Final Exam

Find the power absorbed by the device.

(+4V across the element, 2A flowing into the + terminal)

- a. -8 W
- b. None of all
- ✓ c. 8 W
- d. 4 W

Solution:

Passive sign convention satisfied (current enters + terminal):

$$P = V \times I = 4 \times 2 = 8 \text{ W (absorbed)}$$

A load draws 10 A when connected to a 240V source.
How much energy does the load consume in 75 seconds?

- a. 24 kJ
- b. 1800 J
- c. 2400 J
- ✓ d. 180 kJ

Solution:

$$P = V \times I = 240 \times 10 = 2400 \text{ W}$$

$$E = P \times t = 2400 \times 75 = 180,000 \text{ J} = 180 \text{ kJ}$$

A source free second order series R-L-C circuit.

If $R = 1\text{k}\Omega$, $L = 5\text{H}$ and $C = 2\text{mF}$.

The general natural response $i(t)$ for t will be:

a. $i(t) = e^{-\alpha t}(B_1 \cos \omega_d t + B_2 \sin \omega_d t)$

b. $i(t) = e^{-\alpha t}(A_1 t + A_2)$

✓ c. $i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$

d. none of all

Solution:

$$\alpha = R/(2L) = 1000/10 = 100$$

$$\omega_0 = 1/\sqrt{LC} = 1/\sqrt{5 \times 0.002} = 10$$

$$\alpha > \omega_0 \rightarrow \text{OVERDAMPED}$$

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

